



UNIVERSITÀ
degli STUDI
di CATANIA

DIPARTIMENTO
di ECONOMIA
e IMPRESA



TOKYO 2020



OLYMPIC GAMES UNTIL TOKYO 2020

DATA SCIENCE FOR
MANAGEMENT

DATA BASE AND BIG DATA
ANALYSIS

- Dataset: Kaggle
- More Information:
<https://olympics.com/en/olympic-games/tokyo-2020>

Final Project Report –
Analysis of Olympic data
from 1972 to 2020

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Starting Dataset

This report analyses data of the Olympics games from 1896-2020

Tokyo Summer Olympics from the data source:

<https://www.kaggle.com/datasets/fearsomejockey/olympics-dataset-2020-tokyo-dataset>

The reason for selection of this dataset is because it consists of 10 attributes which would help in visualization, and I can perform a variety of tasks on this dataset.

The data contains parameter such as Name, Sex, Age, Team, Game, Year, Season, Sport and Medal.

The Medal column is a categorial column with values and references as:

0 - No Medal

1 - Bronze Medal

2 - Silver Medal

3 - Gold Medal

This Dataset contains data from different other datasets Mixed for better visualization.

Field Name	Type	Description
Index	Num	Number of People
Name	String	Name of the Athlete
Sex	String	Gender of the Athlete (Male/Female)
Age	Number	Age of the Athlete
Team	String	Team of the Athlete

Games	String	Game the Athlete participated in
Year	Number	Year the athlete Participated in
Season	String	Season (Summer/Winter)
Sport	String	Sport the Athlete Participated in
Medal	number	0- no Medal 1- Bronze Medal 2- Silver Medal 3- Gold Medal

Business Questions:

- Relationship between Male to Female participants?
- Is there any relationship between Gender and Number of Medal won ?
- What is the Minimum, Maximum, and Average Age of Participants?
- Is there any relationship between Age and Number of Medal won?
- Which Country has the highest number of Medal?
- Which Country has the highest number of Gold Medal?
- Which Country has the highest number of Silver Medal?
- Which Country has the highest number of Bronze Medal?
- Which Country has the highest number of female medalists?
- Which Participant has the highest number of Medals?
- Which Participant has the highest number of Gold Medal?
- What are the top 10 Sports based on number of medals?

In the Dataset chapter we analyze the structure of our datasets and in ETL chapter we discuss how data was cleaned, treated and

manipulated in order to obtain the final dataset used for the dashboard creation.

ETL (Extract, Transform and Load)

Introduction and overall ETL structure:

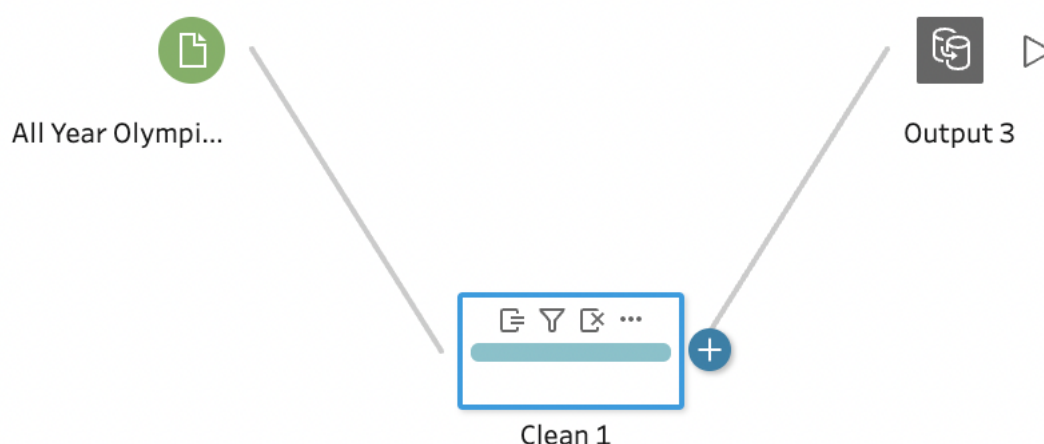
Tableau prep is an **ETL** tool that allows us to extract data from variety of sources, transform the data, and then output the data to a Tableau Data Extract for analysis.

Now before proceeding with the analysis, a preparation phase is mandatory, our data must be cleaned as it needs some changes to be done.

Step 1 Input:

In *Tableau prep* we load our data to start working on it.

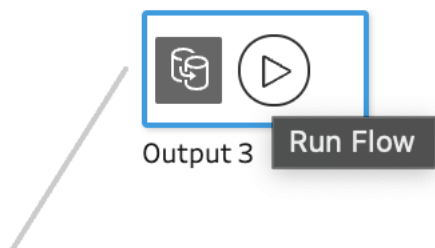
Step 2 Cleaning:



In this step I started to clean and modify my data, the following are the changes I made for better visualization:-

- Changed the name of “**F1**” to “**Index-ID**”
- Merged the Duplicate values (Male = (Male + M), Female = (Female + F)) as the **Sex** column had duplicate variables and changed the attribute name to **Gender**.
- Excluded the values which had null in their **age** column.
- Changed the name of **team** variable to **Country** and merged the teams belonging to same countries which gave the column geographical property as the data will now also be visualized in the world map.
- Short listed the data by only selecting the ‘years’ after 1972 in the **Year** column.

Step 3 Output:



In this step we run the flow and extract our data so that we can work on tableau desktop.

Once the above step is completed our data is then cleaned and we can carry on to the visualization/report part.

Dimensional Fact Model (DFM)

The dimensional fact model (DFM) is an ad hoc and graphical formalism designed to aid in the conceptual modeling phase of the Data Warehouse. It is incredibly user-friendly and can be utilized by both analysts and non-technical people.

As a result, the DFM is a graphical conceptual model created particularly for multidimensional design to:

- lend effective support to conceptual design.
- create an environment in which user queries may be formulated intuitively.
- make communication possible between designers and end users with the goal of formalizing requirement specifications.
- build a stable platform for logical design.
- provide clear and expressive design documentation.

The conceptual representation generated by the DFM consists of a set of fact schemata. Fact schemata model facts, measures, dimensions, and hierarchies. Besides these basic elements, the DFM includes a large set of constructs for expressing the multitude of conceptual nuances that characterize actual modeling scenarios in projects of small to large complexity

A *fact* is a concept relevant to decision-making processes. It typically models a set of events taking place within a company. The fact is represented by a box that displays the fact name along with the measure names.

A *measure* is a numerical property of a fact that describes a quantitative attribute that is relevant to analysis. These are additive measures, can be aggregated over all dimensions using SUM or averaged over all dimensions using AVG.

A *dimension* is a property, with a finite domain, that describes an analysis coordinate of the fact. A fact generally has multiple dimensions that define its minimum representation granularity.

Dimensions are represented by small circles, which are linked to the fact by straight lines.

A *dimensional attribute* is a property, with a finite domain, of a dimension. Like dimensions, a dimensional attribute is represented by a circle.

The relationships among the dimensional attributes are expressed by hierarchies.

A *hierarchy* is a directed tree whose nodes are dimensional attributes and whose arcs model many-to-one associations between dimensional attribute pairs. A hierarchy includes a dimension, positioned at the tree's root, and all of the dimensional attributes that describe it. Arcs are graphically represented by straight lines that connect dimensional attributes. Hierarchies define the way elemental business events can be selected and aggregated for decision-making processes.

In my database there was no need of DFM as I don't have multiple datasets to have a defined hierarchy.

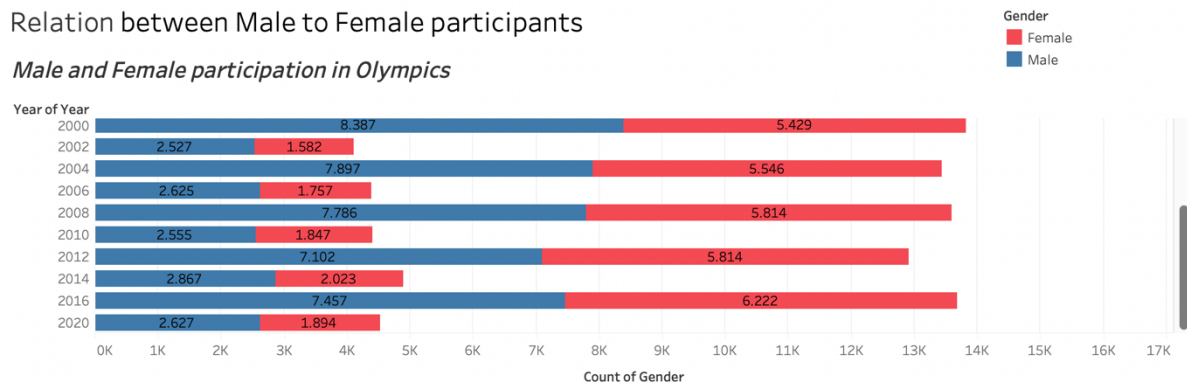
Dashboards

In order to answer the proposed business questions, different dashboards were built for each query, and each containing different information according to the business questions that we proposed.

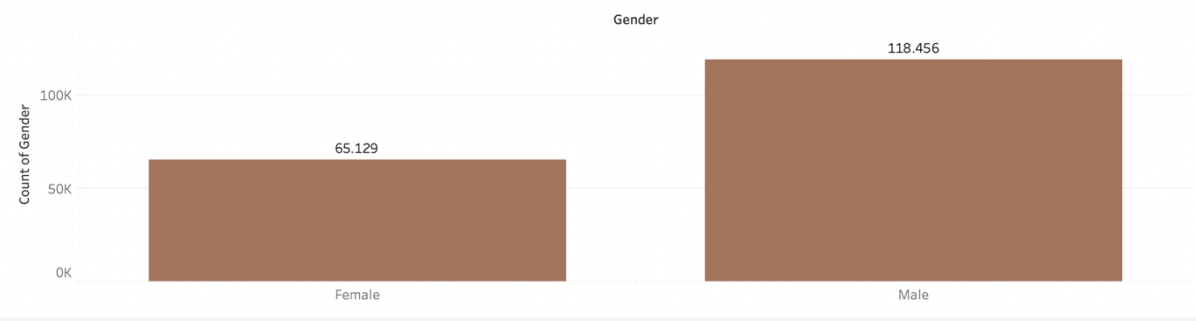
a. Relationship between Male to Female participants?

Relation between Male to Female participants

Male and Female participation in Olympics



Number of Male and Female Participants

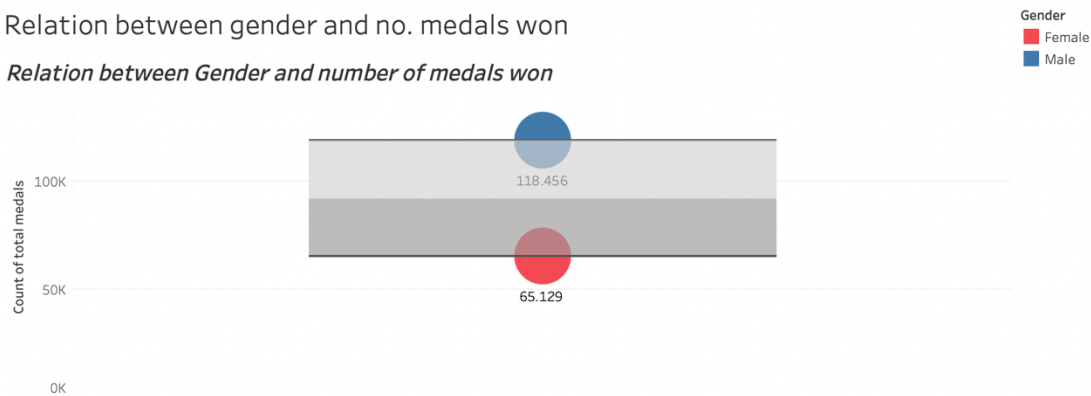


Here we can observe that there is more Male participation than Female in Olympic games. Since the year 1972 to 2020 we can see that according to our data we have only '65,129' number of female athletes whereas male athletes are higher in number specifically '118,456' male athletes competed throughout the time-period.

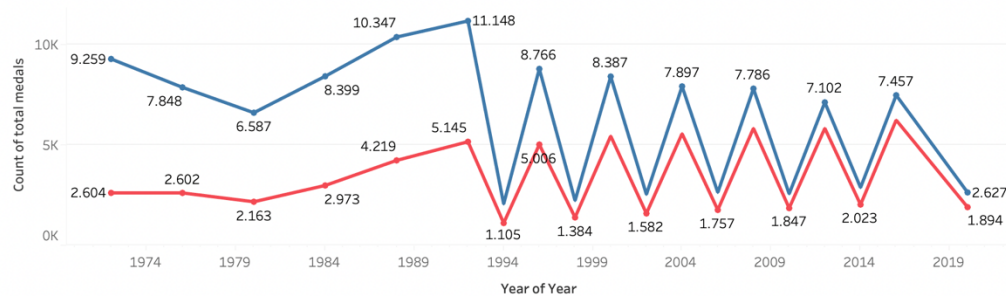
b. Is there any relationship between Gender and Number of Medal won ?

Relation between gender and no. medals won

Relation between Gender and number of medals won



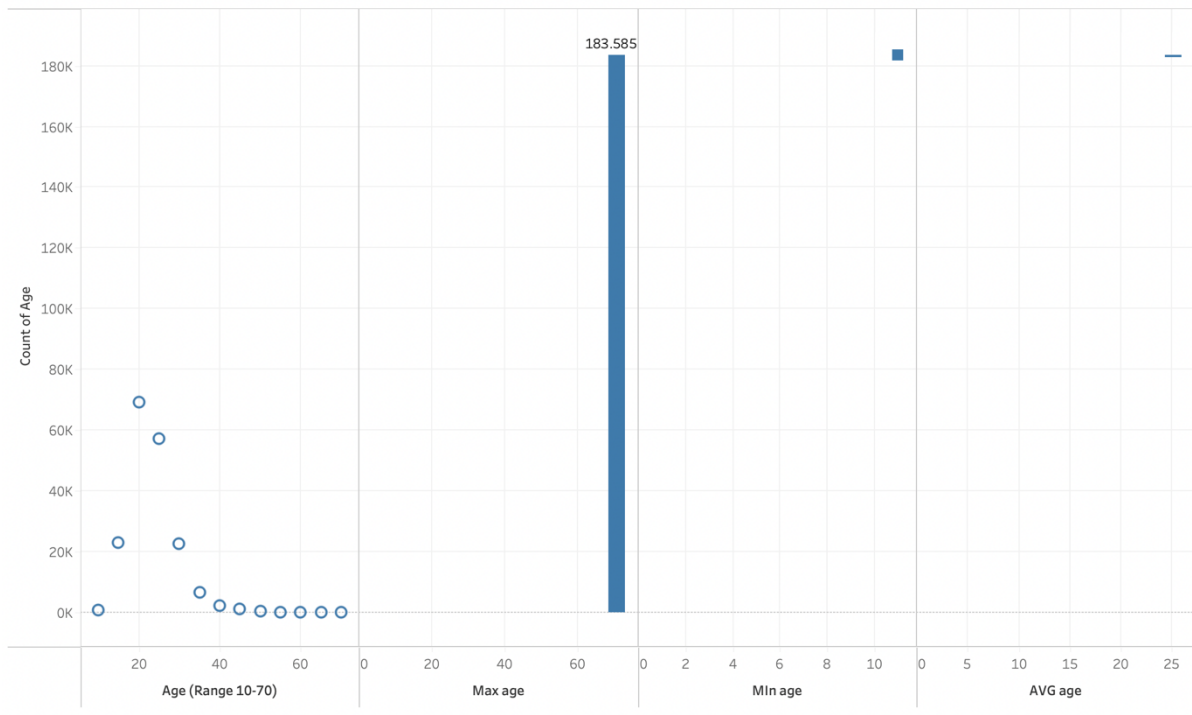
Relation Between Gender and Number of Medals Won(By Year)



We can observe from the above Dashboard the Number of Medals won by both the Genders throughout the time-period of data. We can see also here that the male athletes have won more medals throughout the time.

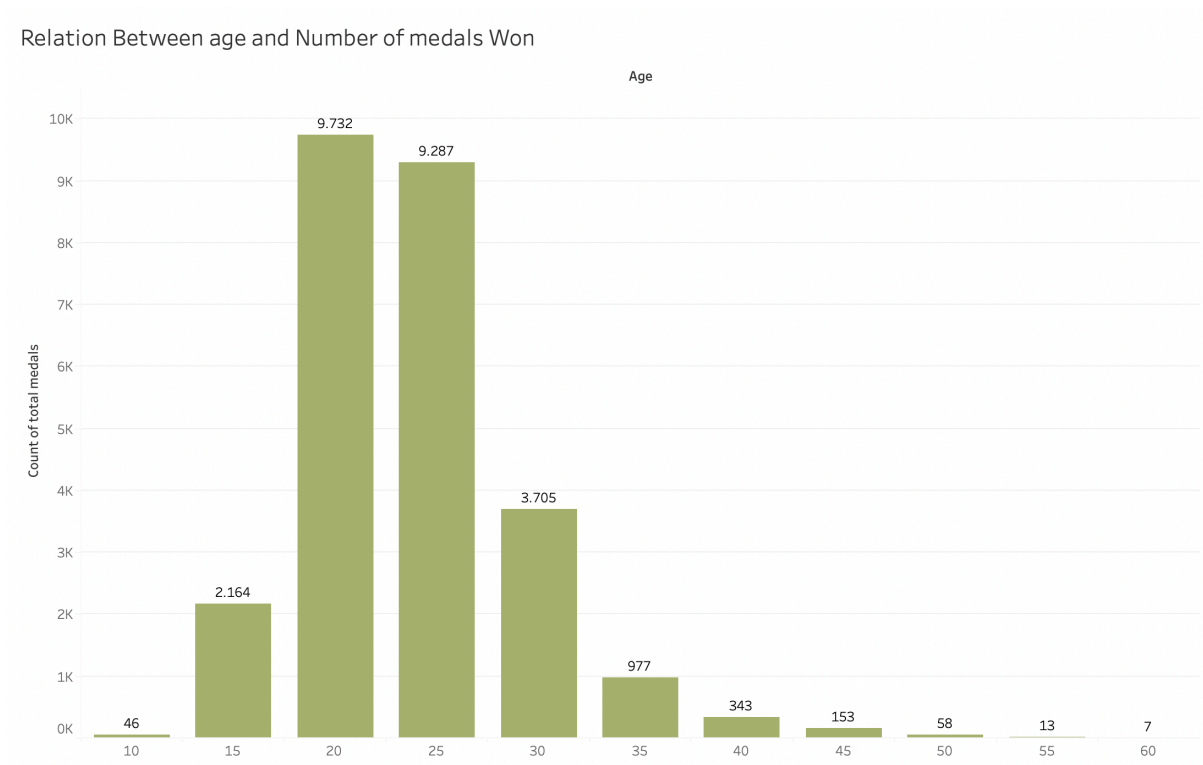
c. What is the Minimum, Maximum, and Average Age of Participants?

Min, Max and Avg of Age of the participants



Here we can observe that most of the Athletes are aged between 20-30 and in the adjacent graph we can observe the max age for all the participants is about 71 and following the graphs the min age is 11 and the average age is 25.16 .

d. Is there any relationship between Age and Number of Medal won?



Here we can observe the Relation Between Age and number of Medals Won. As the bar-graph suggests that the maximum number of athletes who have won medals are aged 20-30.

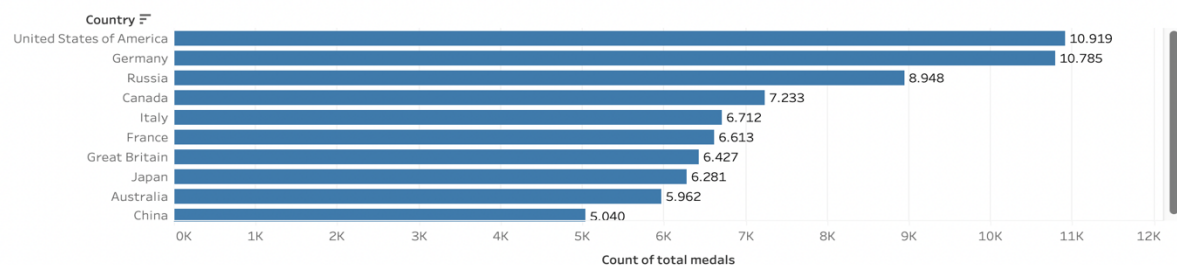
e. Which Country has the highest number of Medal?

Country with the Highest Number of Medal

All the Countries with the number of Medals



Top 10 countries according to no. of medals



In the above Dashboard we can see the Countries with the maximum number of medals. Also, we can find out the top 10 countries with the most medals in sequence.

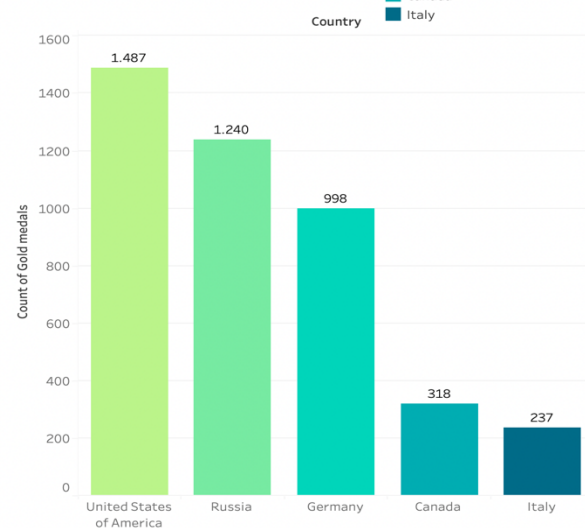
f. Which Country has the highest number of Gold Medal?

Country With Highest Number of Gold medal

Number of Gold Medals



Top 5 Countries With Gold Medal

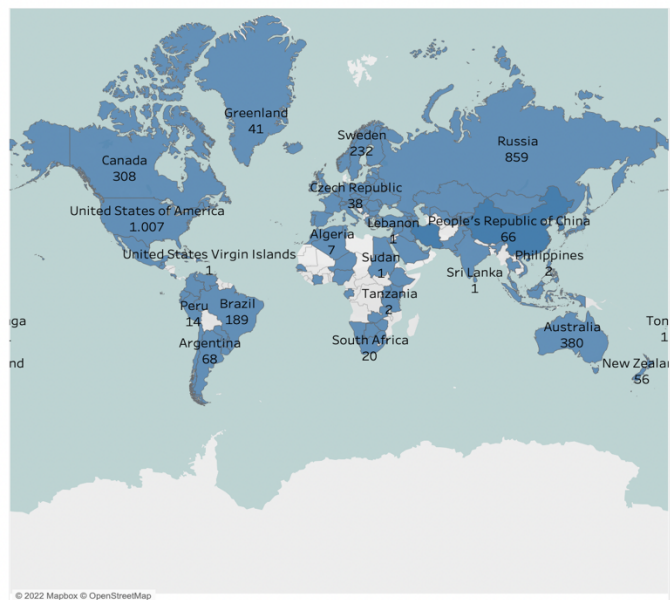


In this Dashboard we are finding out the Countries with highest amount of Gold Medals.

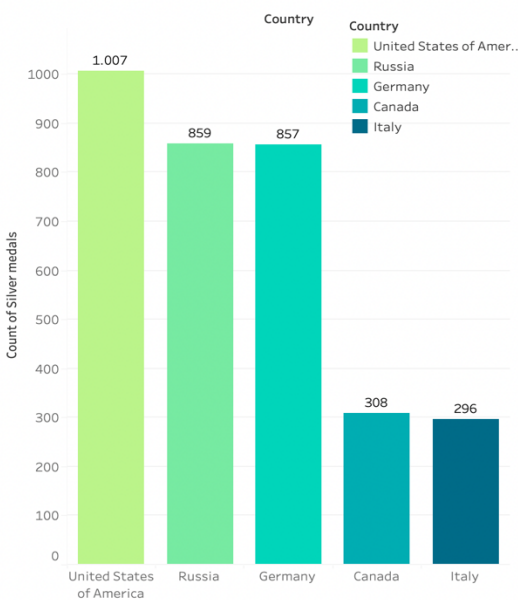
g. Which Country has the highest number of Silver Medal?

Country with Highest Number of Silver Medal

Number of Silver Medals



Top 5 countries with Silver Medal

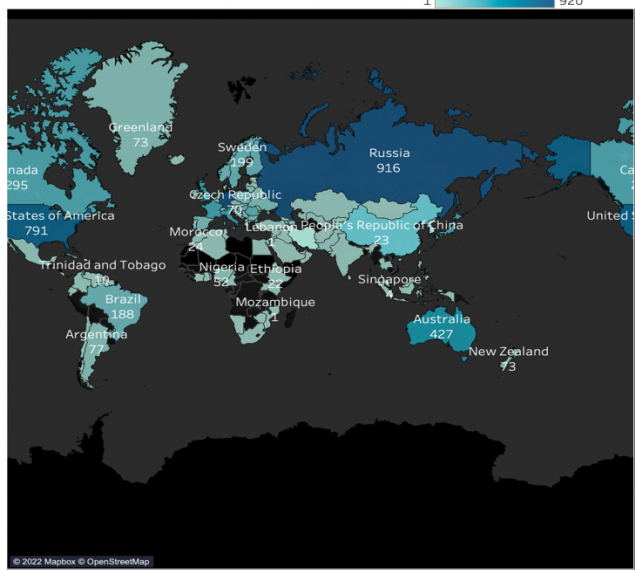


In this Dashboard the Countries with Silver medals are shown in Ranking. We can also observe the top 5 countries.

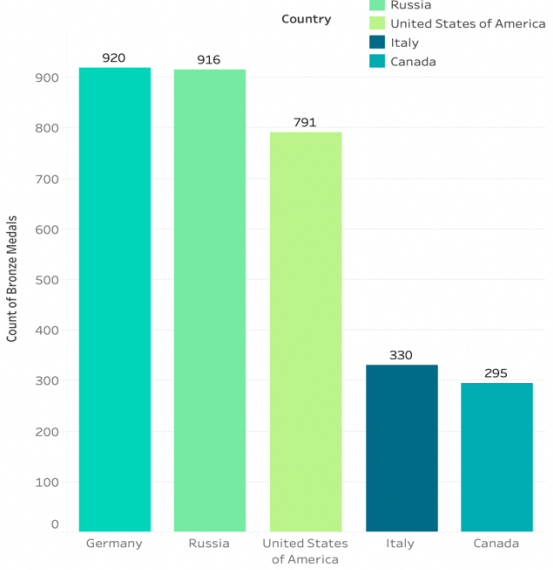
h. Which Country has the highest number of Bronze Medal?

Countries with highest Bronze medal

Number of Bronze Medals



Top 5 countries with Bronze Medal

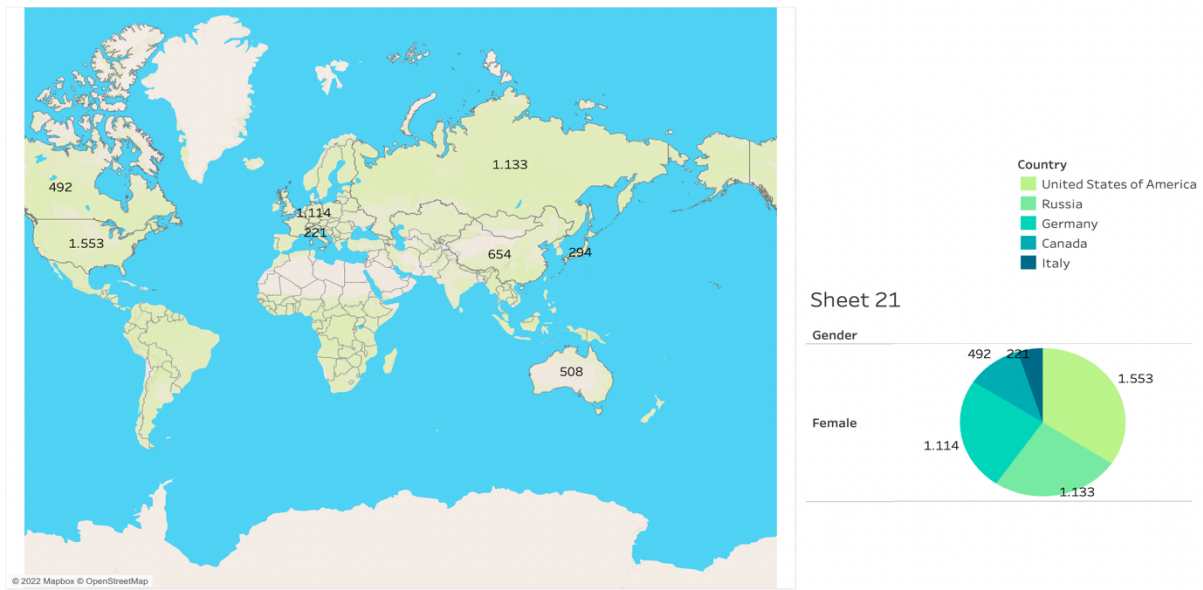


In this Dashboard we are observing the countries with the maximum bronze medals only also the Ranking of the countries in descending order.

I. Which Country has the highest number of female medalists?

Countries with High Number of Female Athletes

Country with the Highest Female Medalists

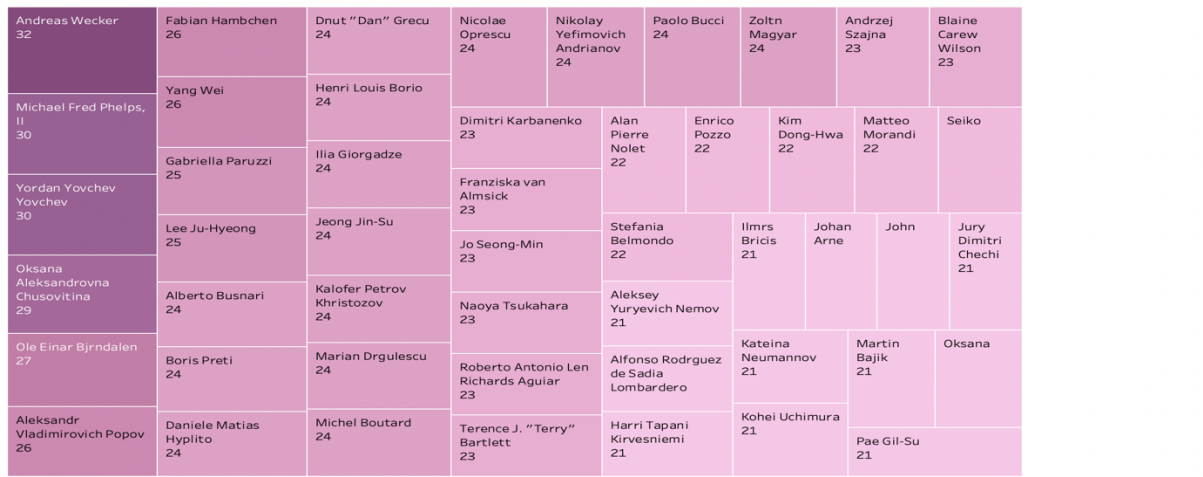


Here we are only finding out the Country ranking of only medals won by female athletes. We can also see on the map (only the top 5 countries).

j. Which Participant has the highest number of Medals?

Participant with the highest number of medals

Players with most Medals



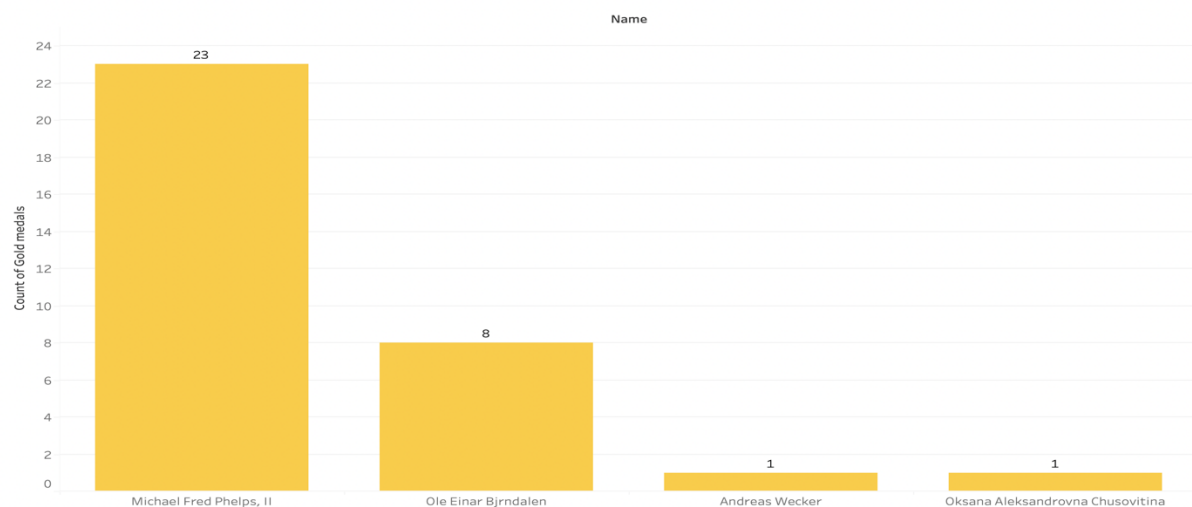
Here we can observe the Athlete with the highest number of Medals throughout the years of our Data.

We can see that the Athletes with more medals are highlighted by darker color.

k. Which Participant has the highest number of Gold Medal?

Player with highest number of Gold Medals

Top 4 Athletes based on the number of Gold Medals

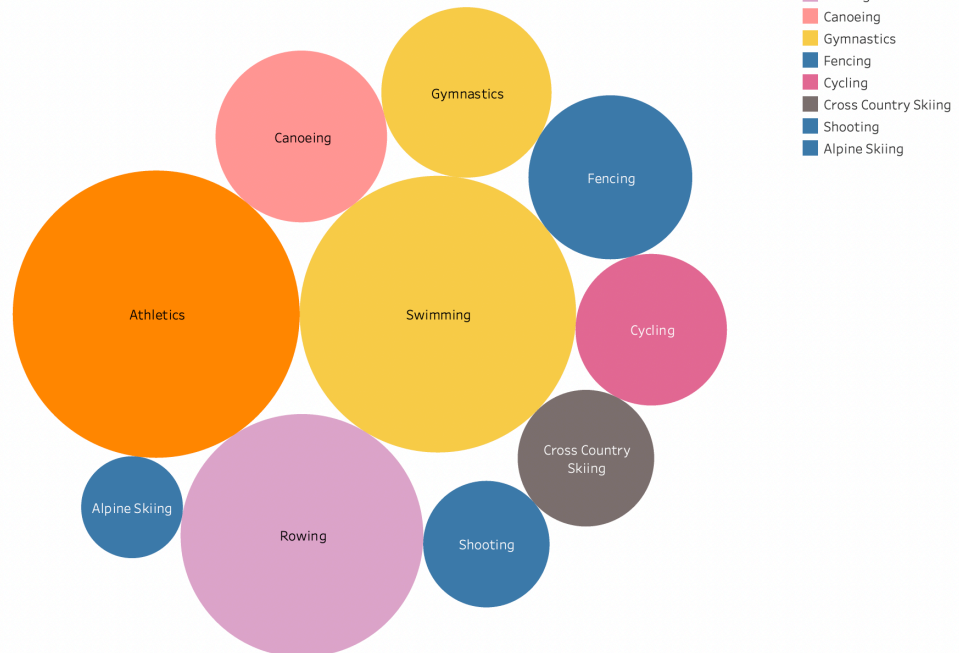


In this Dashboard we are observing only the athletes with high number of gold medals.

I. What are the top 10 Sports based on number of medals?

Top 10 sports based on number of medals

Top 10 Sports with most medals



Here we can find through the analysis of data the top 10 sports in Olympics based on the number of medals won throughout the years.

Conclusion and Summary

1. Based on the data we can see that in Olympics the participation of male athletes is relatively higher than female athletes. Which would be because of many factors as to personal choices or Country's situation with favoritism of a gender.
2. We can also observe that Male athletes tend to win more medals in the games as compared to women.
3. 20-30 years of Age is the ideal for people to Win a Medal in the Olympics also the most common age of the Athletes.
4. We can also observe that USA, Russia and Germany are the top 3 countries dominating the world in terms of Medals won.
5. We found that the Player with Maximum number of medal is Andreas Wecker with 32 medals in total and the player with maximum number of gold medals is Michael Fred Phelps II.
6. Also the sports in which people tend to win more medals are:
 - a. Athletics
 - b. Swimming
 - c. Rowing

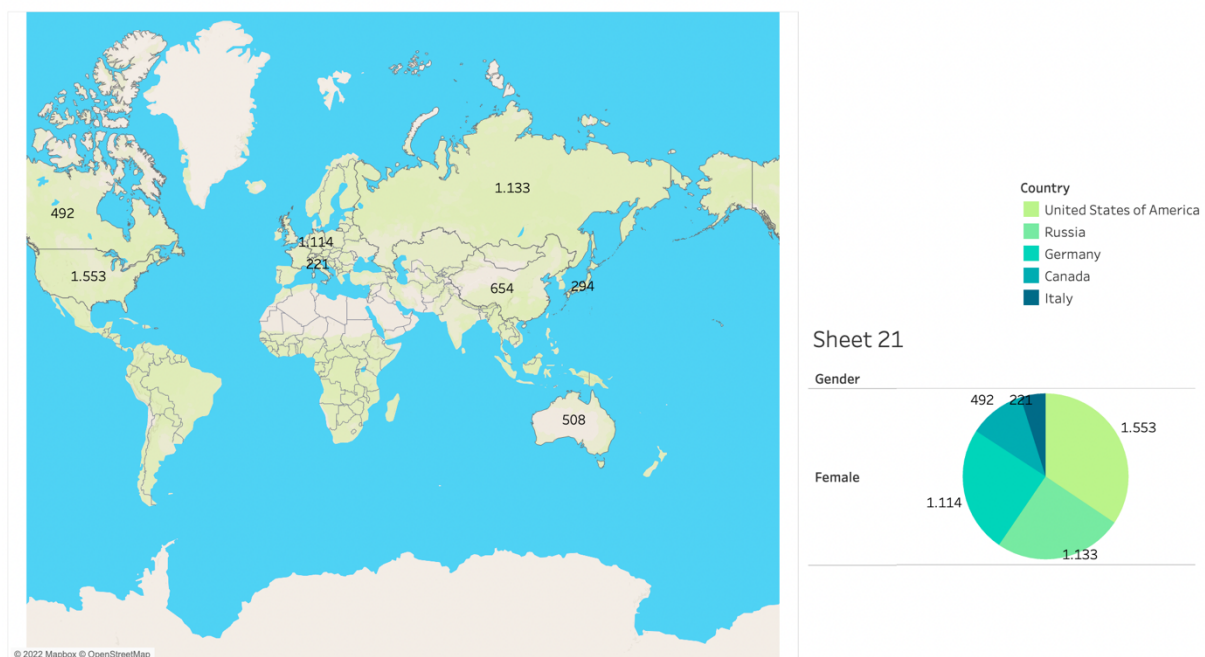
Color Blindness Test

The Color Blindness Test measures the Ability to tell the difference between Colors.

My normal view of Dashboard:

Countries with High Number of Female Athletes

Country with the Highest Female Medalists

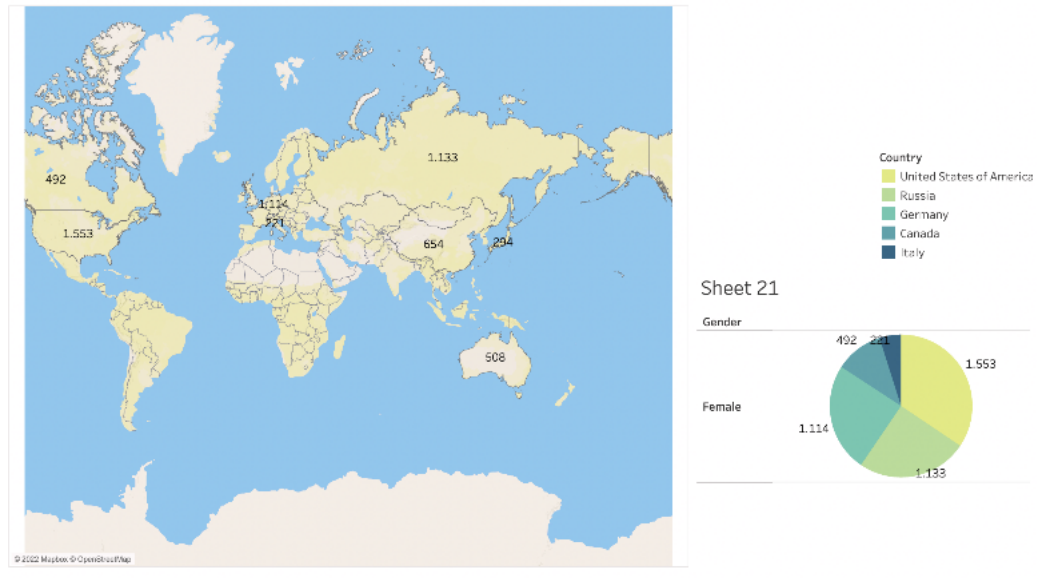


Anomalous Trichromacy:

1) Red-Weak/Protanomaly

Countries with High Number of Female Athletes

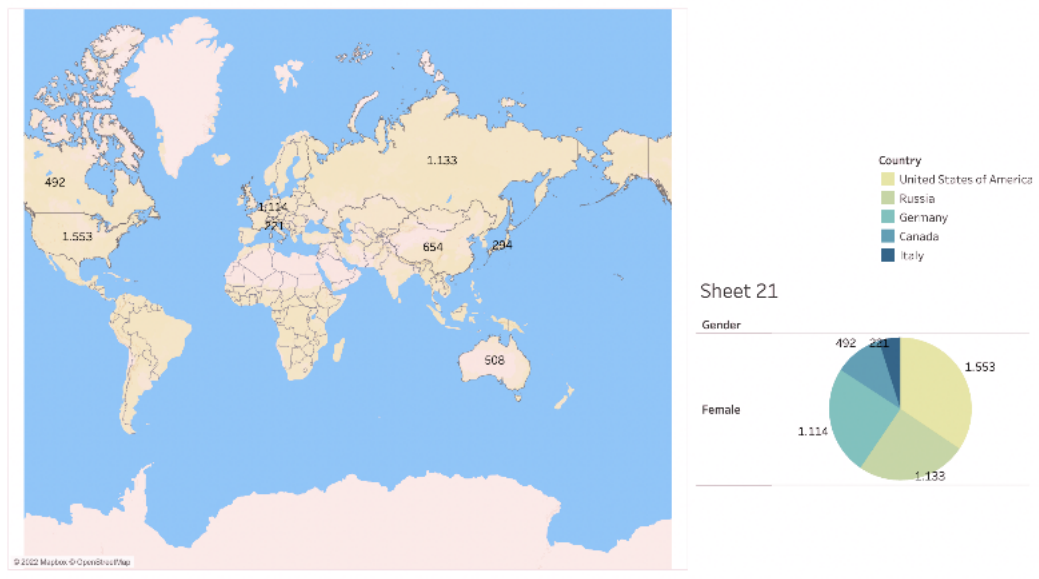
Country with the Highest Female Medalists



2) Green-weak/Deuteranomaly

Countries with High Number of Female Athletes

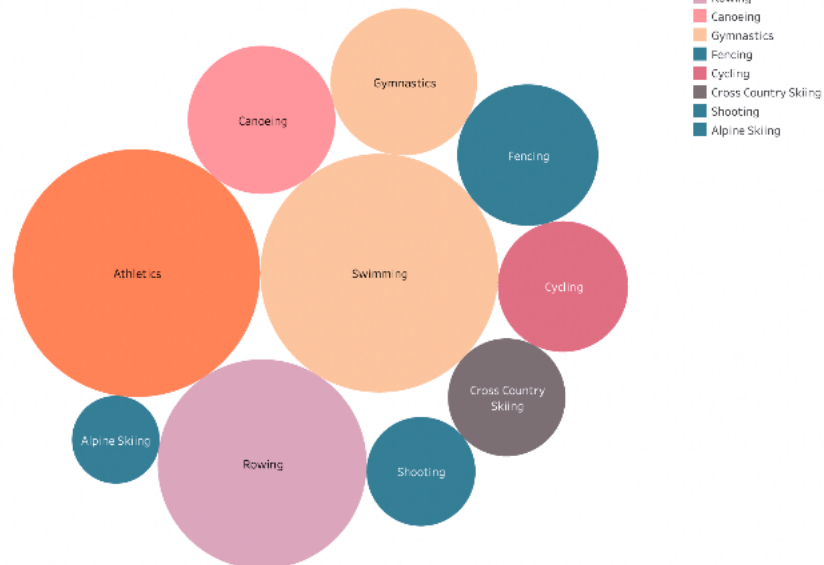
Country with the Highest Female Medalists



3) Blue-weak/Tritanomaly

Top 10 sports based on number of medals

Top 10 Sports with most medals

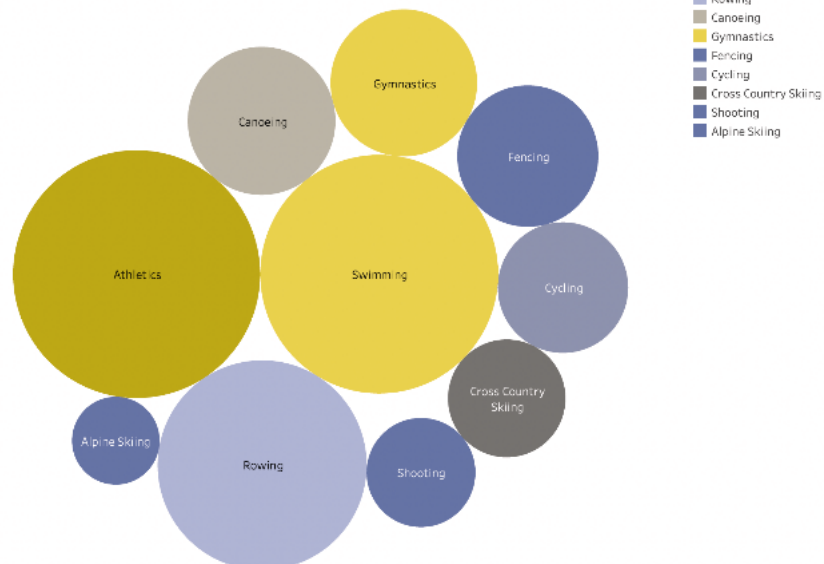


Dichromatic View:

1) Red-Blind/Protanopia

Top 10 sports based on number of medals

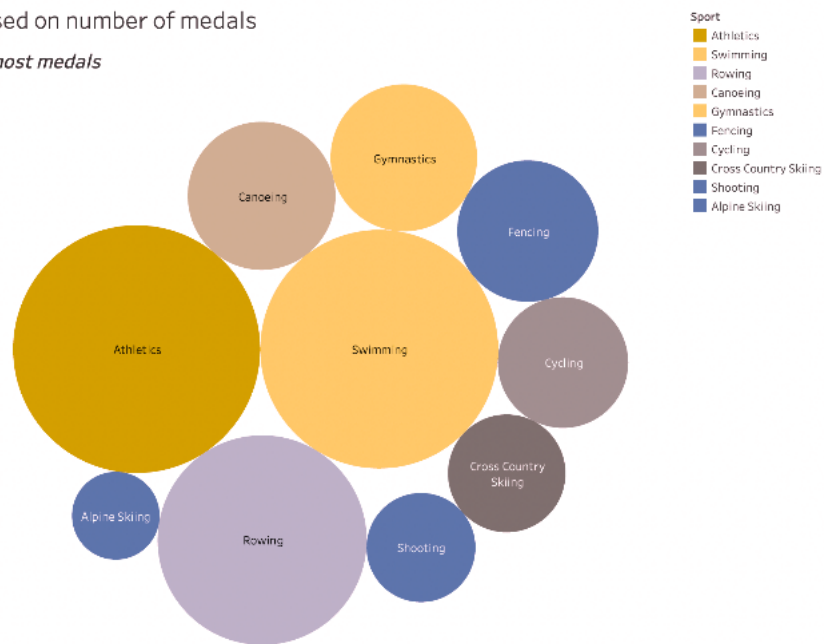
Top 10 Sports with most medals



2) Green-blind/Deuteranopia

Top 10 sports based on number of medals

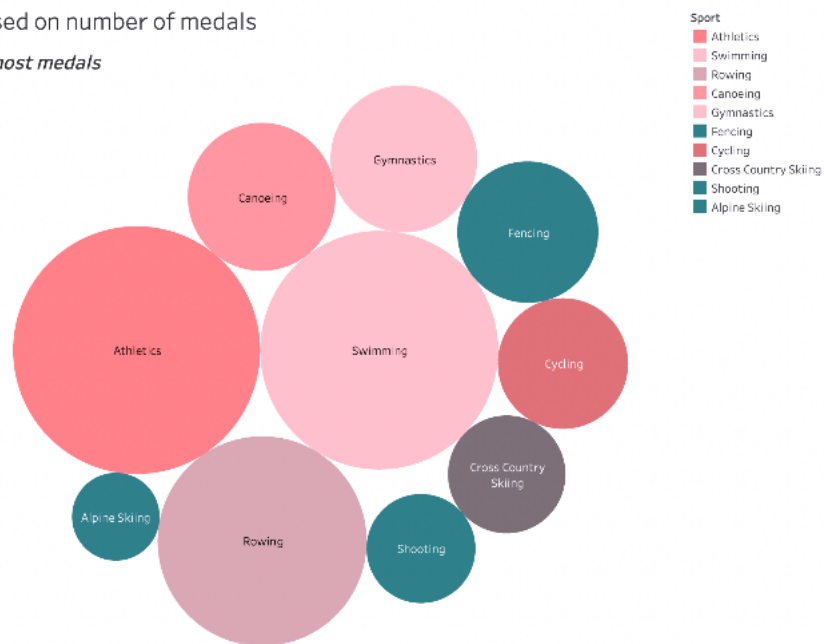
Top 10 Sports with most medals



3) Blue-blind/Tritanopia

Top 10 sports based on number of medals

Top 10 Sports with most medals

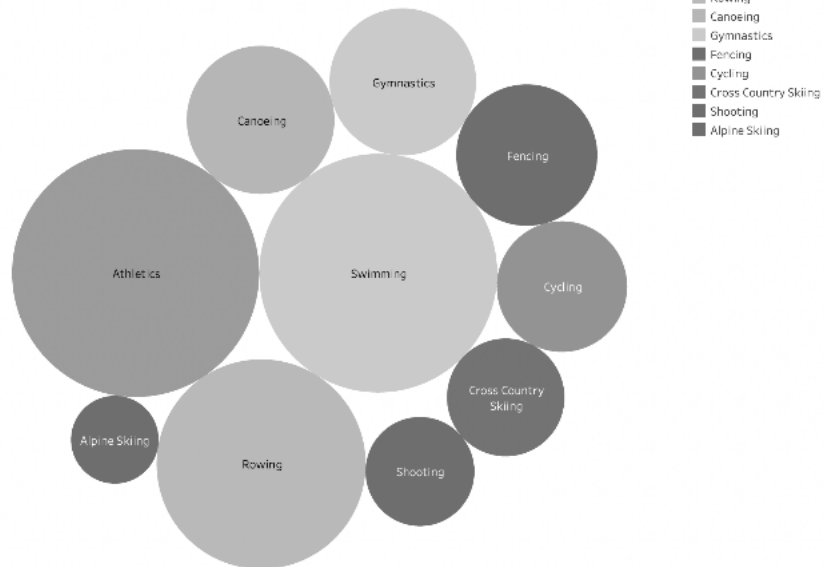


Monochromatic View:

1) Monochromacy

Top 10 sports based on number of medals

Top 10 Sports with most medals



2) Blue cone Monochromacy

Top 10 sports based on number of medals

Top 10 Sports with most medals

